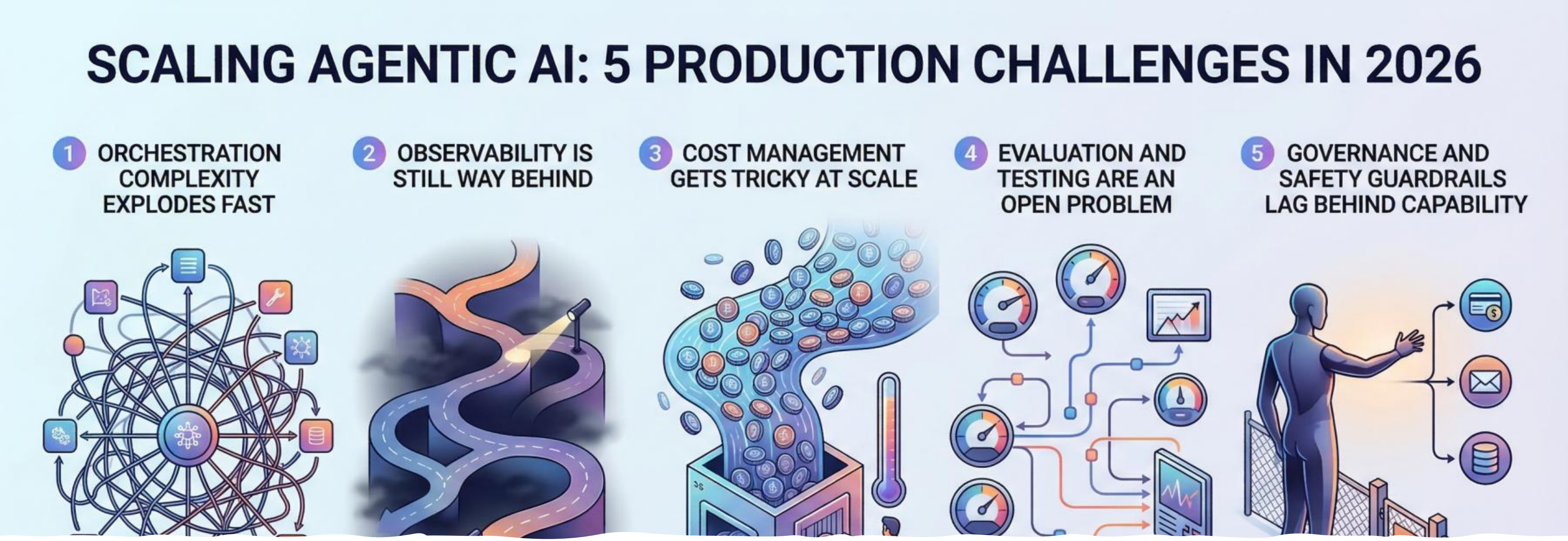
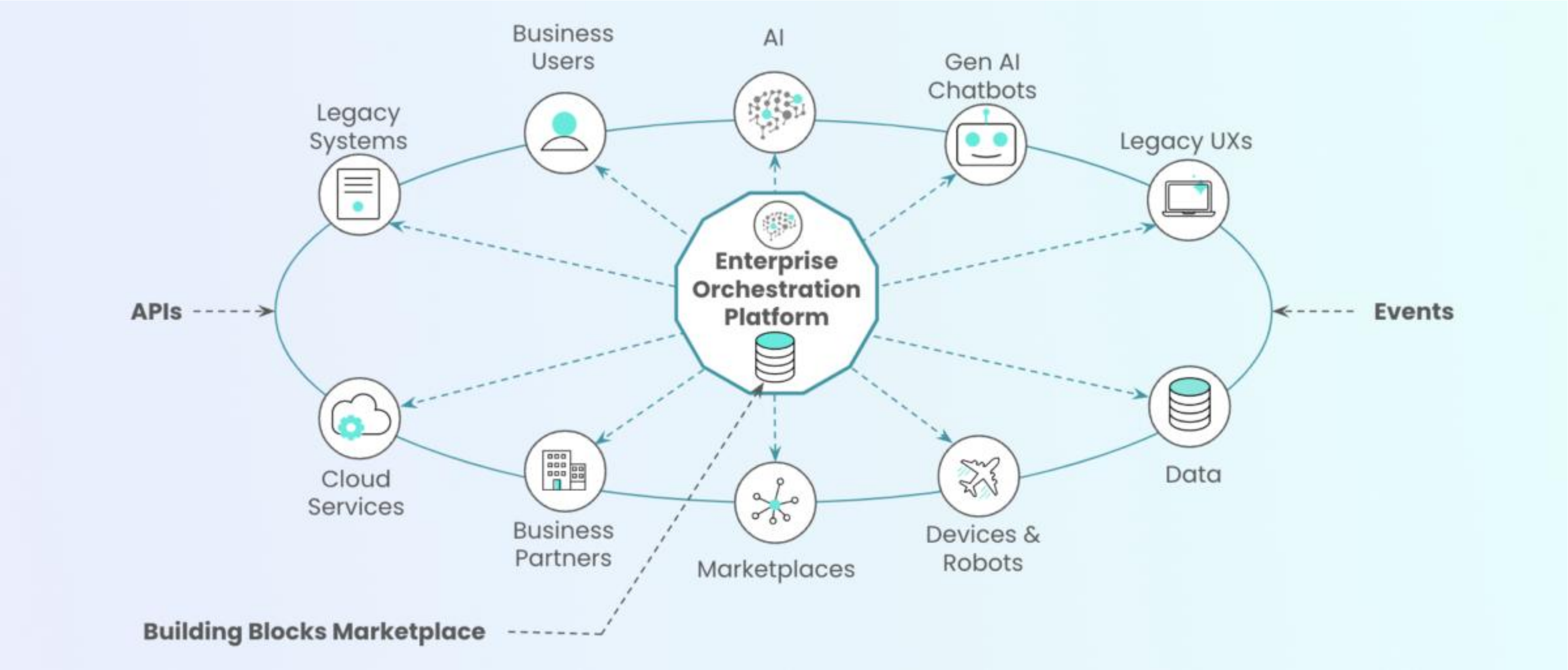


The Grand Capstone Challenge

- **Goal:** Solve a high-stakes, real-world problem by building an end-to-end autonomous system.
- **The Paradigm Shift:** Moving from "Experimental Code" to a **Mission-Critical Product** that is scalable, observable, and fully containerized.
- **The Framework:** Leveraging **LangGraph** for the core orchestration, as it provides the state management required for complex medical or industrial pipelines.

Technical Requirements (The "Agentic Orchestration")

- **3+ Specialized Agents:** At least one Manager/Auditor agent to enforce quality and self-correction.
- **External Integration:** Real-world "Hands" (e.g., searching live medical databases or controlling IoT sensors).
- **Production Layer:** A FastAPI backend with asynchronous streaming and Pydantic validation.
- **DevOps Layer:** A fully Dockerized deployment with multi-container orchestration.



Capstone Project Ideas

Autonomous Medical Research Pipeline: An agentic team that retrieves patient history via RAG, searches latest medical literature, and drafts a clinical summary.

Industrial IoT Failure Predictor: A system that monitors sensor data (Tools), analyzes patterns (Agents), and generates maintenance alerts through an API.

Wildcard - Automated Regulatory Compliance Agent: An agent that monitors local legal updates and automatically checks if a company's internal PDFs (Memory Vault) are still compliant.

Final Audit & System Defense

Code Review: A deep dive into your LangGraph state logic and error-handling protocols.

Architecture Defense: Justifying your choice of rerankers, tools, and orchestration patterns.

Observability Check: Proving the system is "traceable" and doesn't enter infinite loops in production.